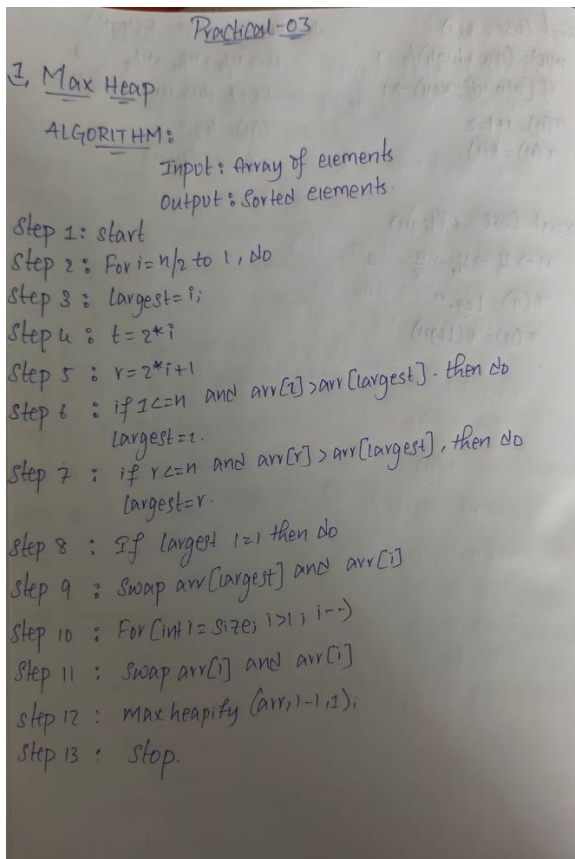


PRACTICAL-03

AIM: Implementation of max-heap sort algorithm.

1.) MAX HEAP

ALGORITHM:



PROGRAM:

```
#include <iostream>

using namespace std;

void maxheapify(int arr[], int size, int i)
{
    int largest = i;
    int l = 2 * i;
    int r = 2 * i + 1;
    if (l <= size && arr[l] > arr[largest])
```

```

{
    largest = l;
}

if (r <= size && arr[r] > arr[largest])
{
    largest = r;
}

if (largest != i)
{
    swap(arr[largest], arr[i]);
    maxheapify(arr, size, largest);
}
}

void heapsort(int arr[], int size)
{
    for (int i = size / 2; i >= 1; i--)
    {
        maxheapify(arr, size, i);
    }
    for (int i = size; i > 1; i--)
    {
        swap(arr[1], arr[i]);
        maxheapify(arr, i - 1, 1);
    }
}

int main()
{
    int n;

    cout << "Enter number of elements: ";
    cin >> n;
    int arr[n + 1];

    cout << "Enter elements: ";
    for (int i = 1; i <= n; i++)
    {

```

```

    cin >> arr[i];

}

heapsort(arr, n);

cout << "Sorted array: ";

for (int i = 1; i <= n; i++)

{

    cout << arr[i] << " ";

}

return 0;

}

```

OUTPUT:

```

Output
Enter number of elements: 5
Enter elements: 5
9
7
6
1
Sorted array: 1 5 6 7 9
=== Code Execution Successful ===

```

TIME COMPLEXITY:

Time Complexity

for $i = n/2, i \geq 1; i--$
 max heapify (arr, n, i)
 for $i = n, i \geq 1; i--$
 swap (arr[i], arr[1])
 maxheapify (arr, i-1, 1)

$$T(n) = n \times \log n + n \times \log n$$

$$= n \log n + n \log n$$

$$T(n) = O(n \log n)$$

Best Case :-

for $i = n/2$ to 1
 max heapify (arr, n, i) $\rightarrow n$
 for $i = n$ to 2
 swap (arr[i], arr[1])
 max heapify (arr, i-1, 1) $\rightarrow \log n$
 $T(n) = O(n) + n \times O(\log n)$
 $T(n) = O(n) + O(n \log n)$
 $T(n) = O(n \log n)$

3. Average Case $\rightarrow n$

for $i = n$ to 2
 swap $(i \rightarrow 1)$
 maxheapify $(1 \rightarrow \log n)$
 $T(n) = n + 1 + \log n$
 $T(n) = O(n) + O(n \log n)$
 $T(n) = O(n \log n)$

3. Worst Case

Max heapify from root to lowest level of the heap
 $\Rightarrow \log_2 n$
 maxheapify $(1 \rightarrow \log n)$
 $T(n) = n + n \times \log n$
 $T(n) = O(n \log n)$

